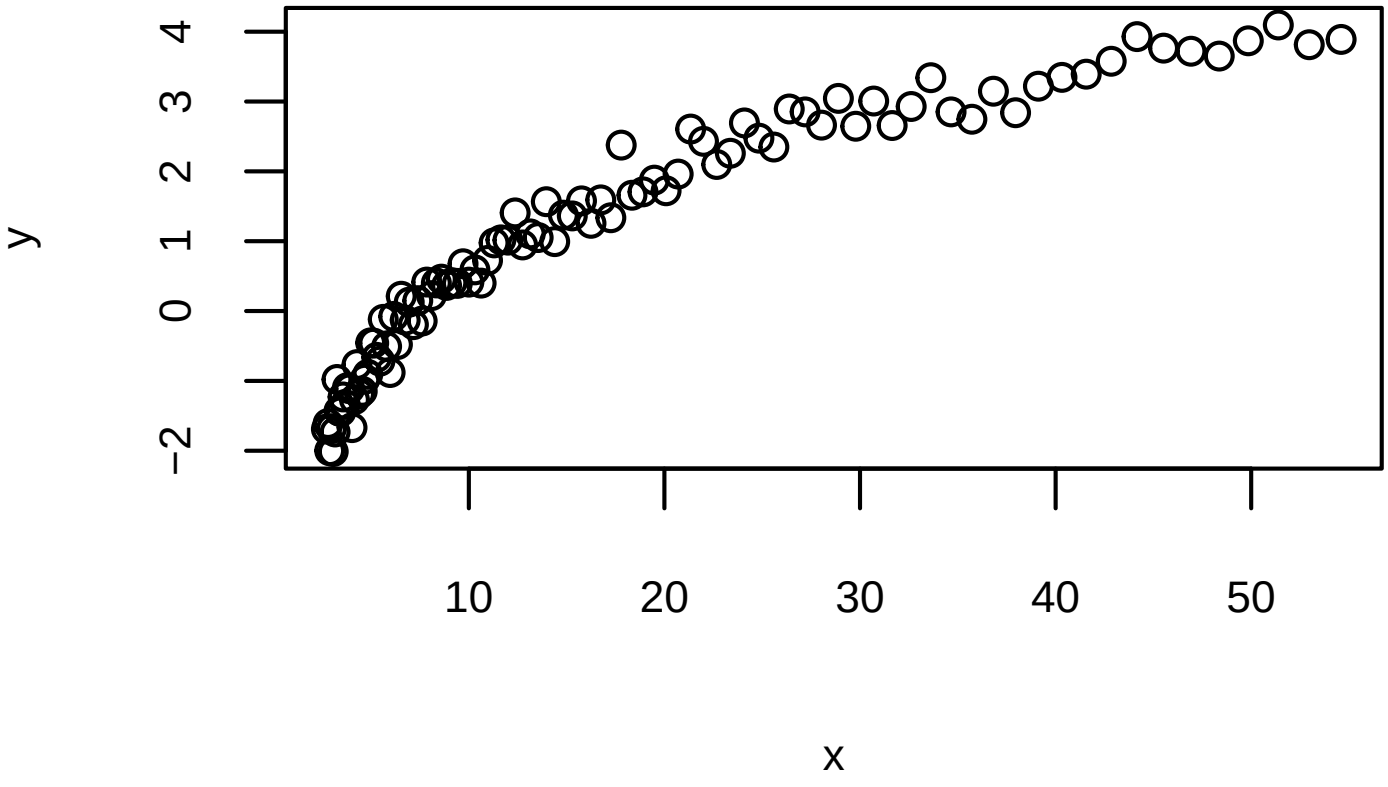
Diagnostics and Remedial Measures

Example 1

Example 1

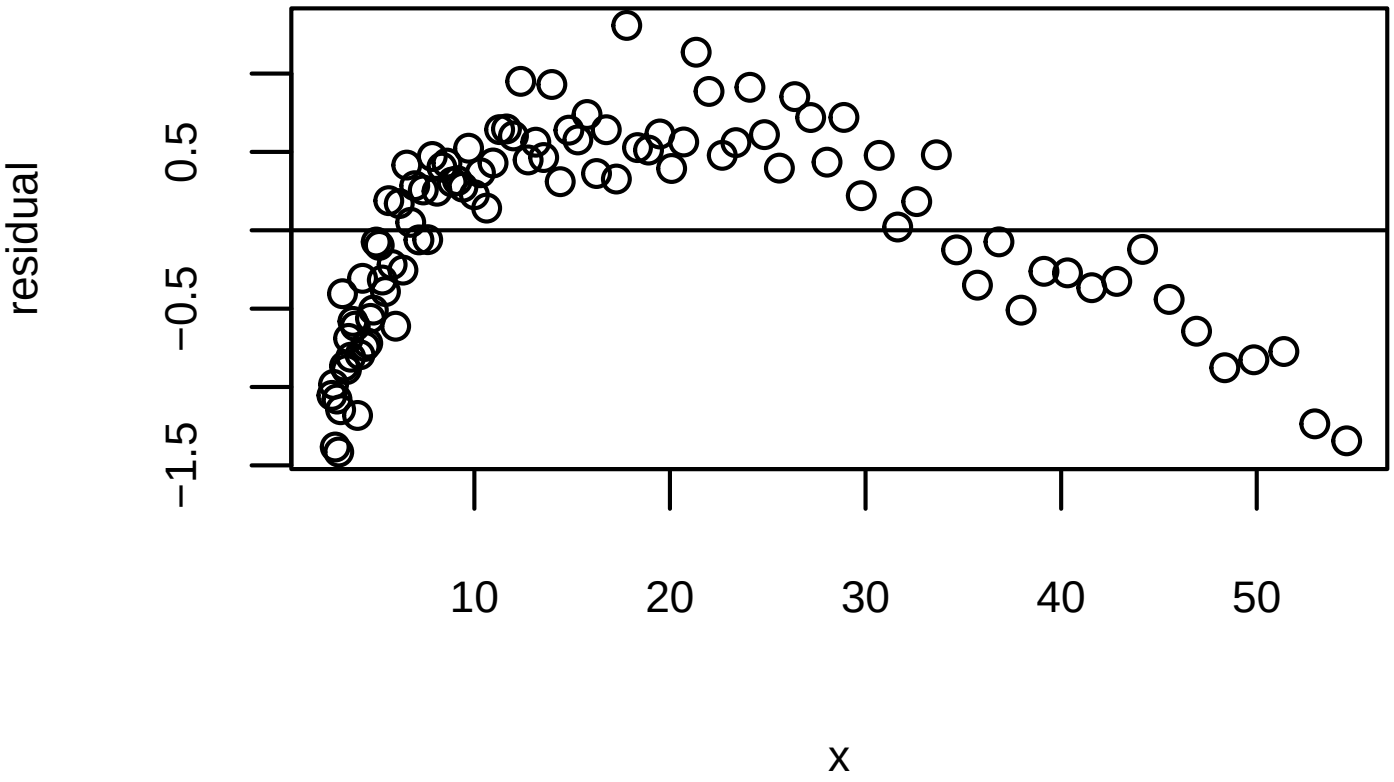
Start by looking at a scatter plot of the data, x and y.



The relationship between x and y appears to be non-linear. It is not obvious whether the y values are more spread out for different values of x.

Example 1

What do the residuals from a fitted linear regression model look like?



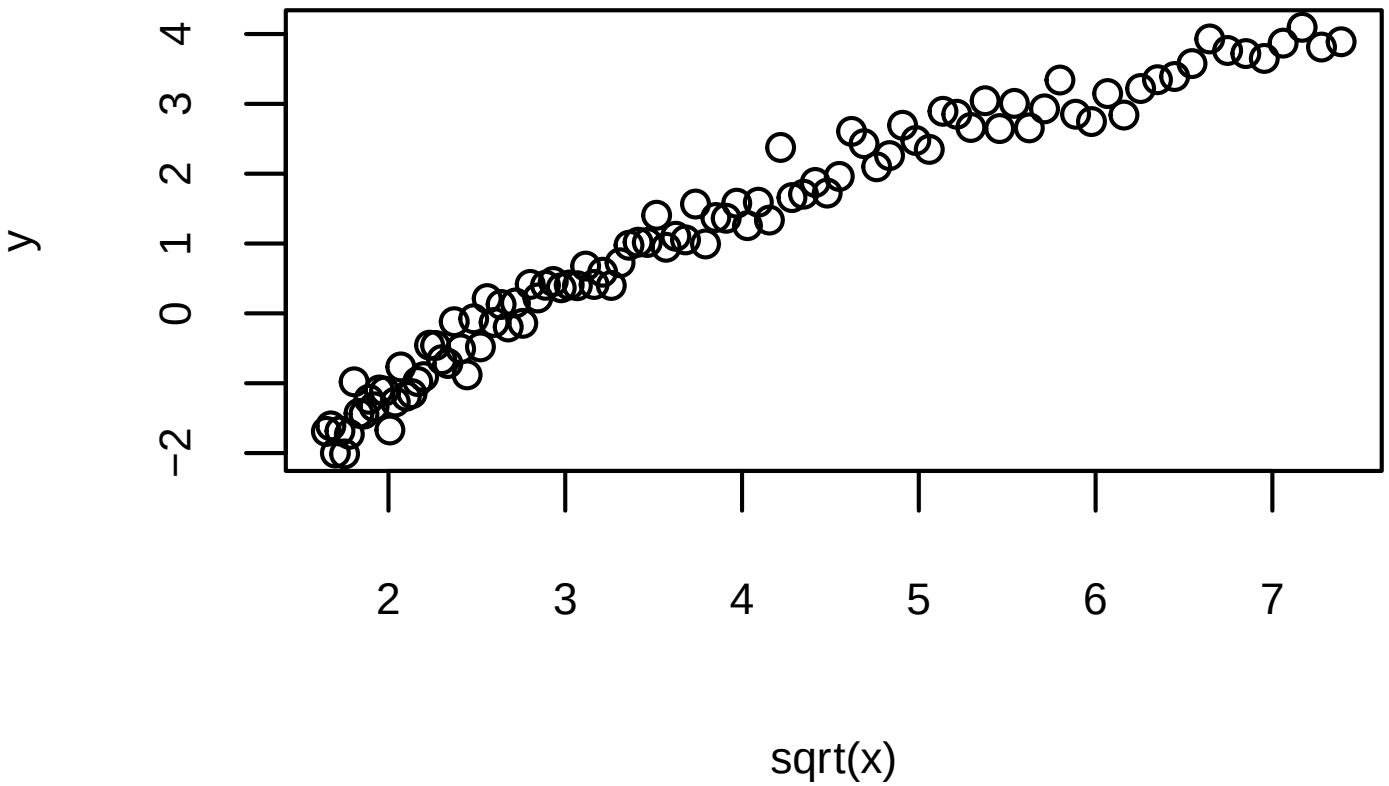
There is a clear pattern in the residuals: they tend to be negative for very small and very large values of x, and positive in the middle.

Both plots suggest that the true relationship is not linear.

Example 1

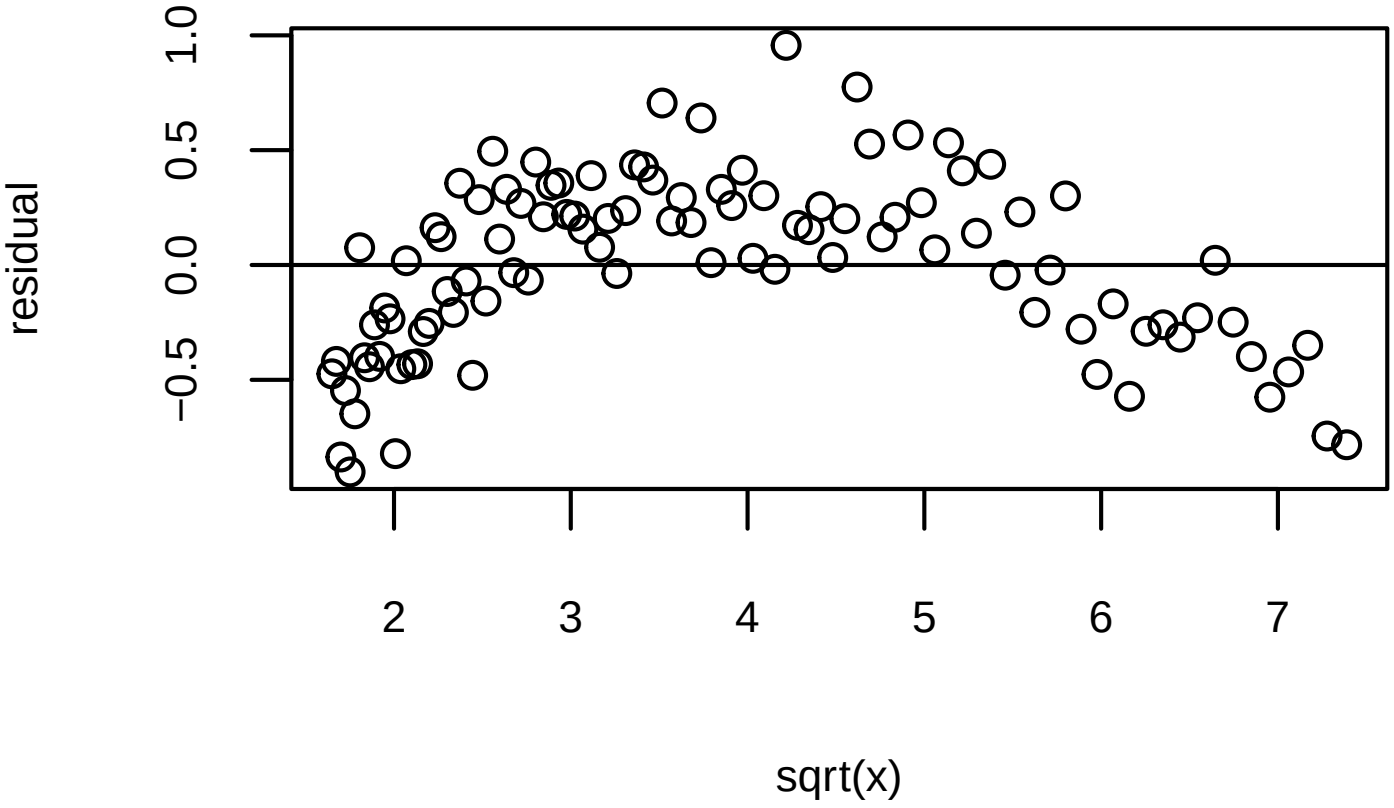
Because we aren't sure if the error terms have constant variance or not, let's try transforming the x variable. (Transforming y changes how spread out the y values are.)

In general, popular transformations include log and square root.



This looks better, but the relationship still looks a bit curved rather than linear.

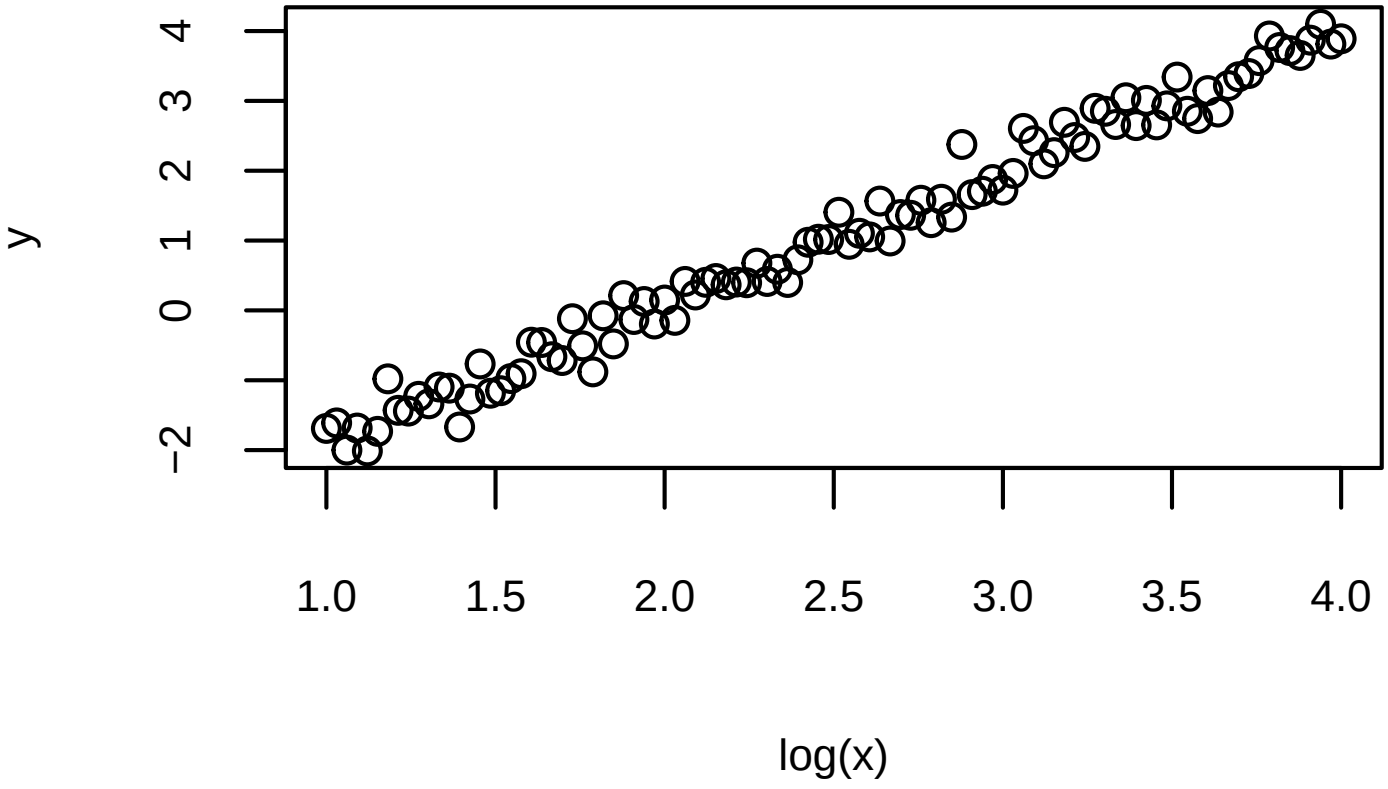
Example 1



If we try fitting a model to (sqrt(x), y), the residuals plot indicates that we haven't fixed the non-linearity problem. The residuals still have an obvious pattern depending on the value of x.

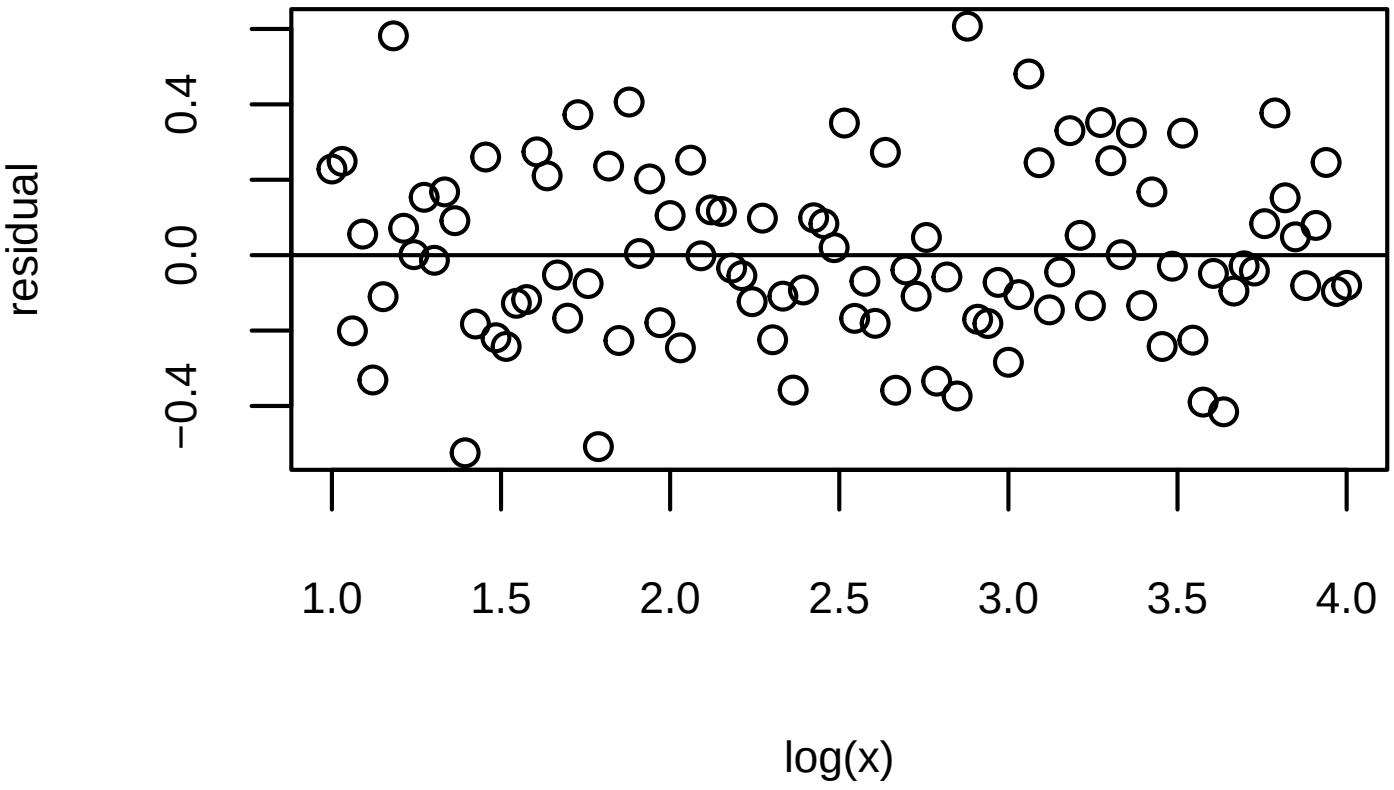
Example 1

We can try a different transformation of x:



The relationship between log(x) and y looks linear. What happens when we fit a linear regression to log(x) and y?

Example 1

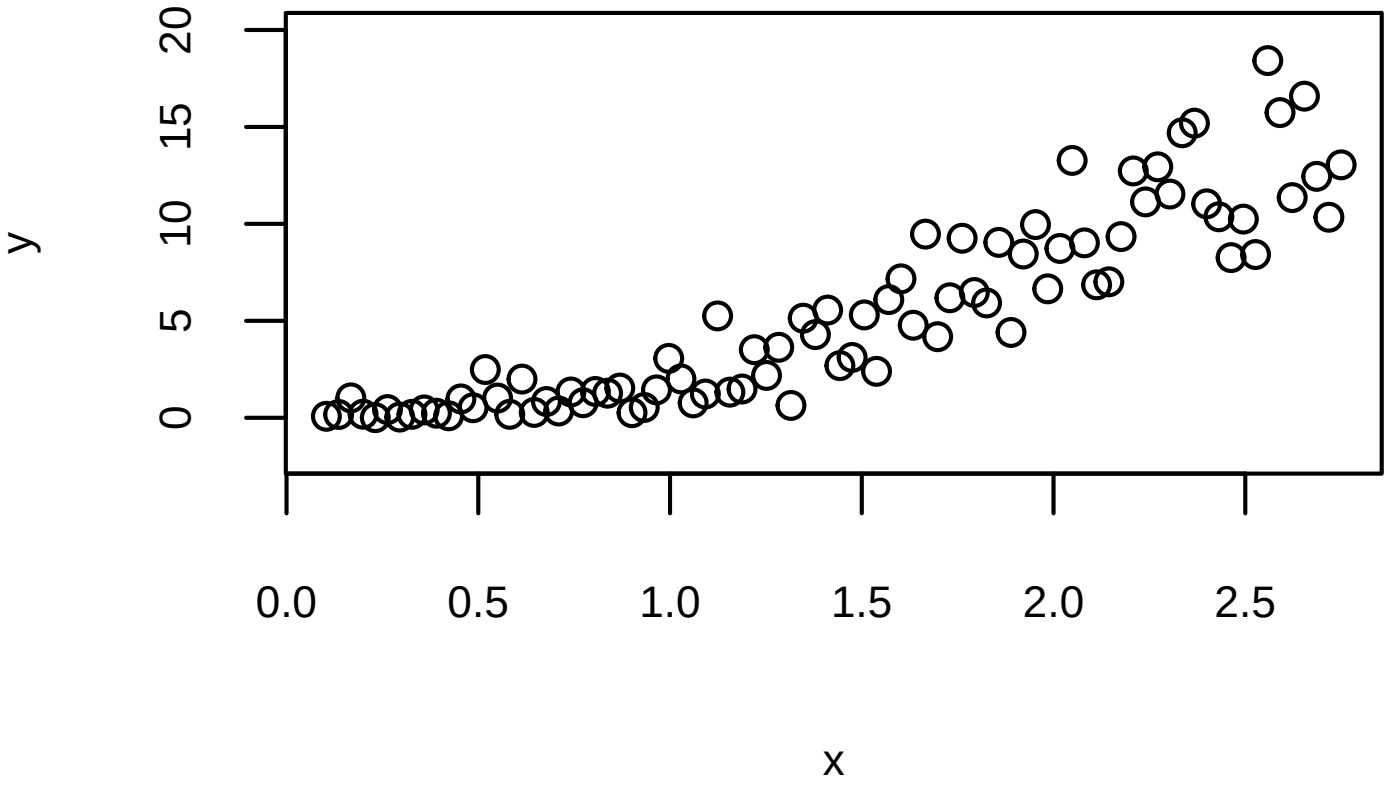


Using log(x) instead of x, the residuals from the fitted linear model look randomly scattered around zero, and appear to have similar variance across different values of log(x). This is satisfactory.

Example 2

Example 2

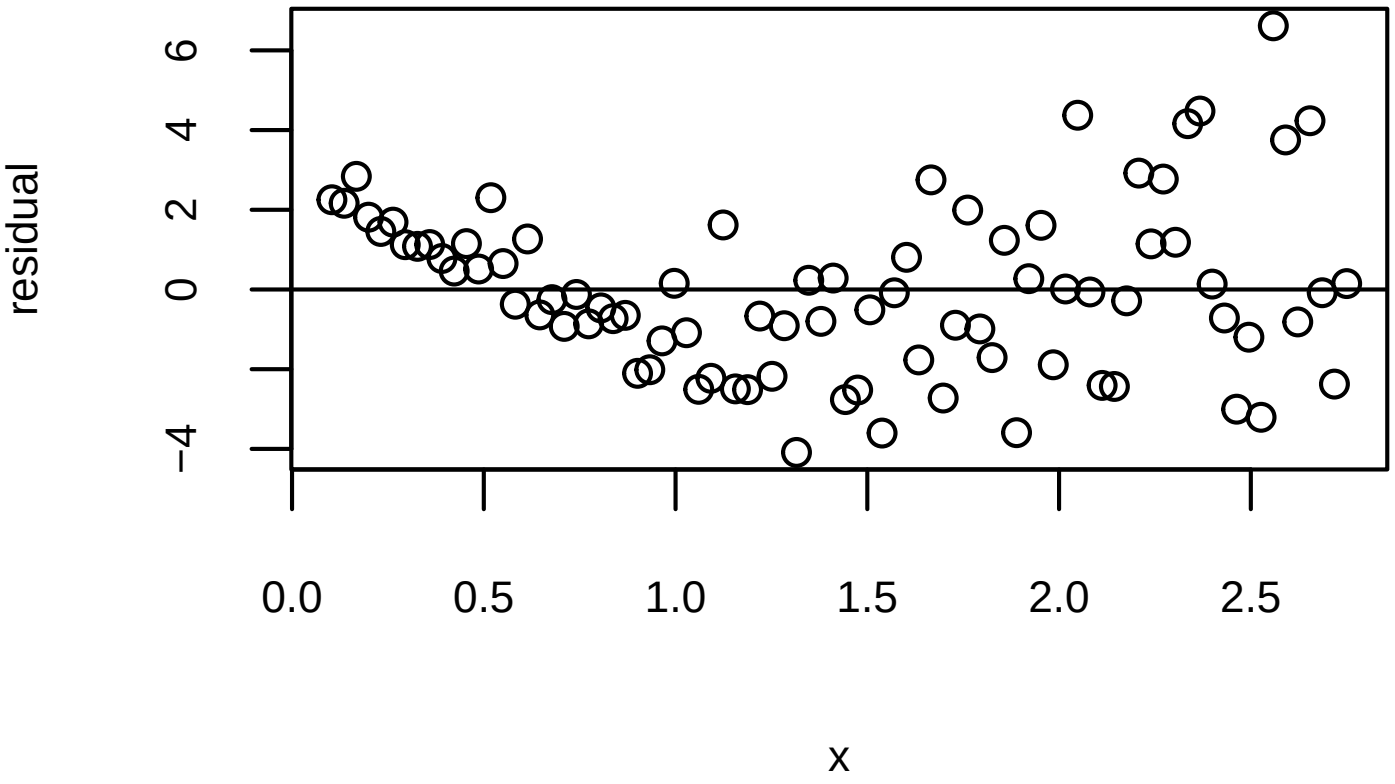
Start by looking at a scatter plot of the data, x and y.



Based on the scatter plot, it looks like the relationship between x and y may be somewhat curved, rather than a straight line. Not extremely clear.

Example 2

Suppose we fit a linear model to this x and y data. What do the residuals from that model look like?



If model assumptions correct: residuals should be scattered evenly above and below zero, across all values of x, with similar magnitude for all values of x.

That does not seem to be the case here.

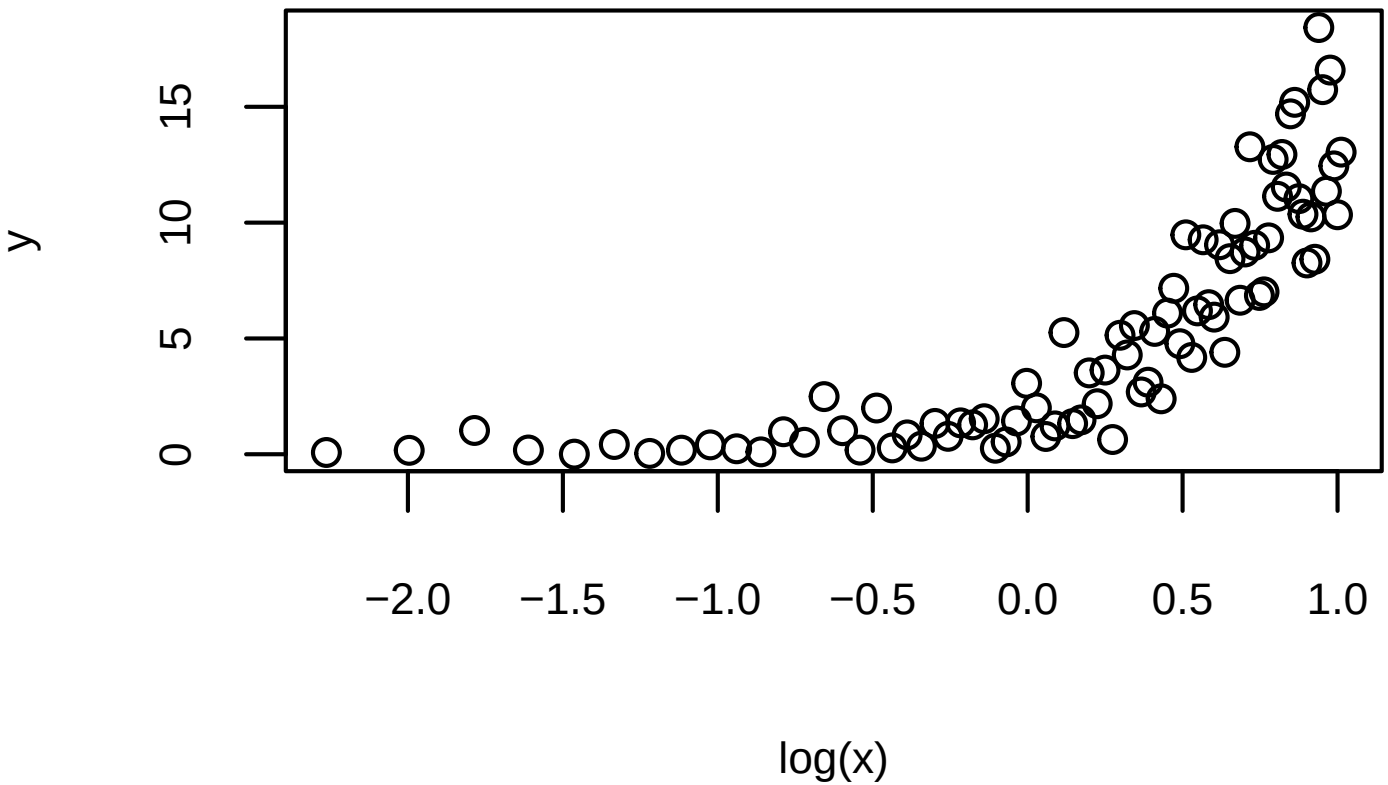
Two potential issues: violation of linearity, and non-constant variance of the error terms.

Example 2

Idea: try transforming x and/or y, and fitting a new model to the transformed variables, to reflect a non-linear relationship.

Popular transformations to try: log (of x, or y, or both), square root/squared

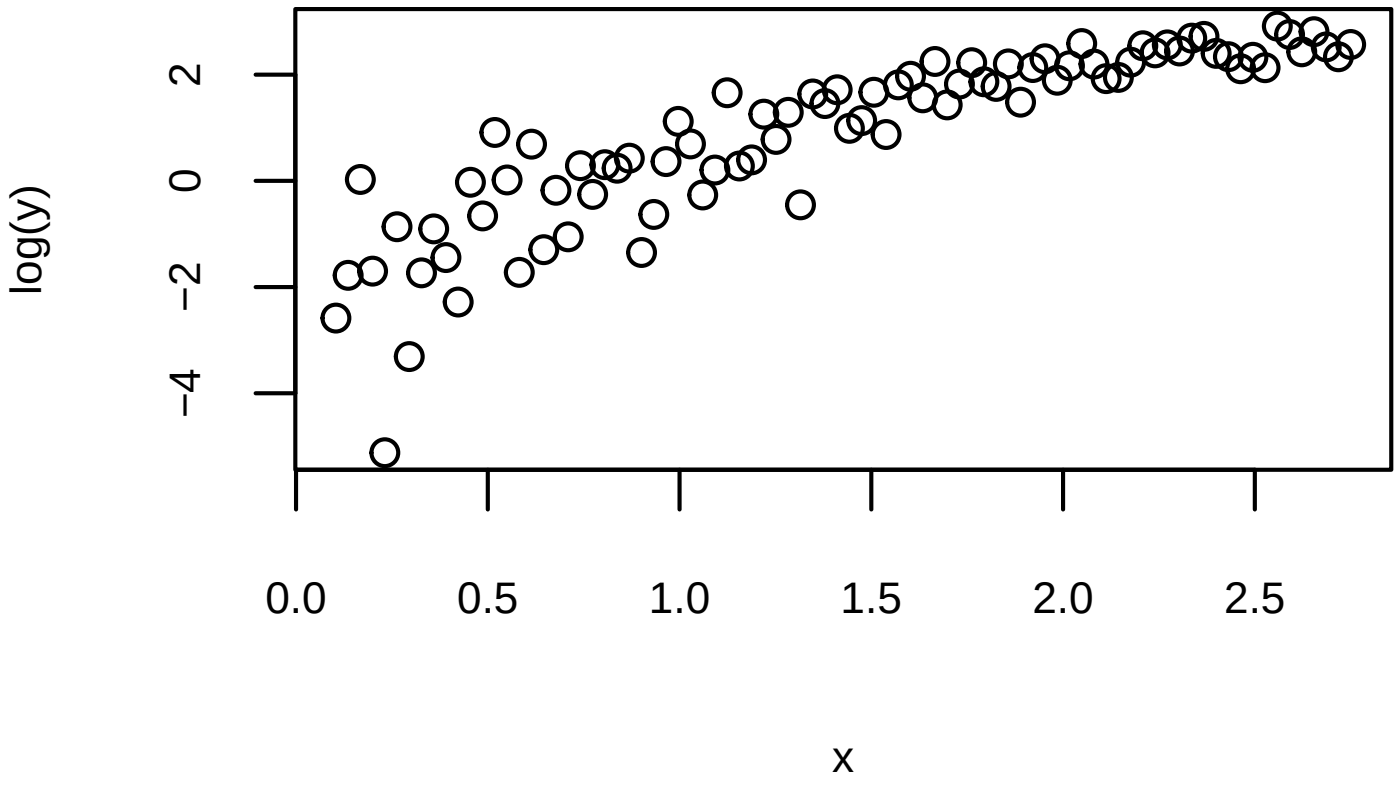
Try taking the log of x:



That didn't help- relationship between log(x) and y is nonlinear.

Example 2

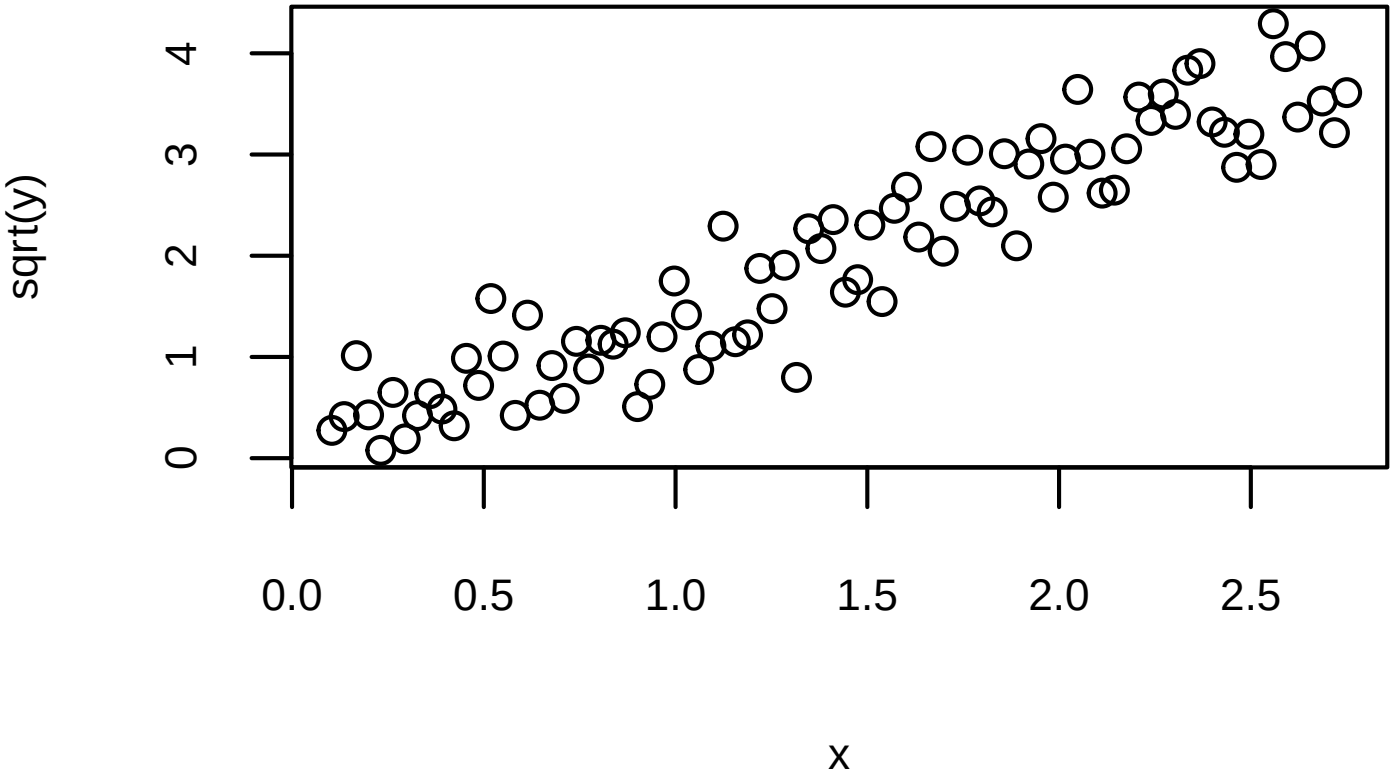
What about log of y?



The relationship between x and log(y) doesn't look linear either.

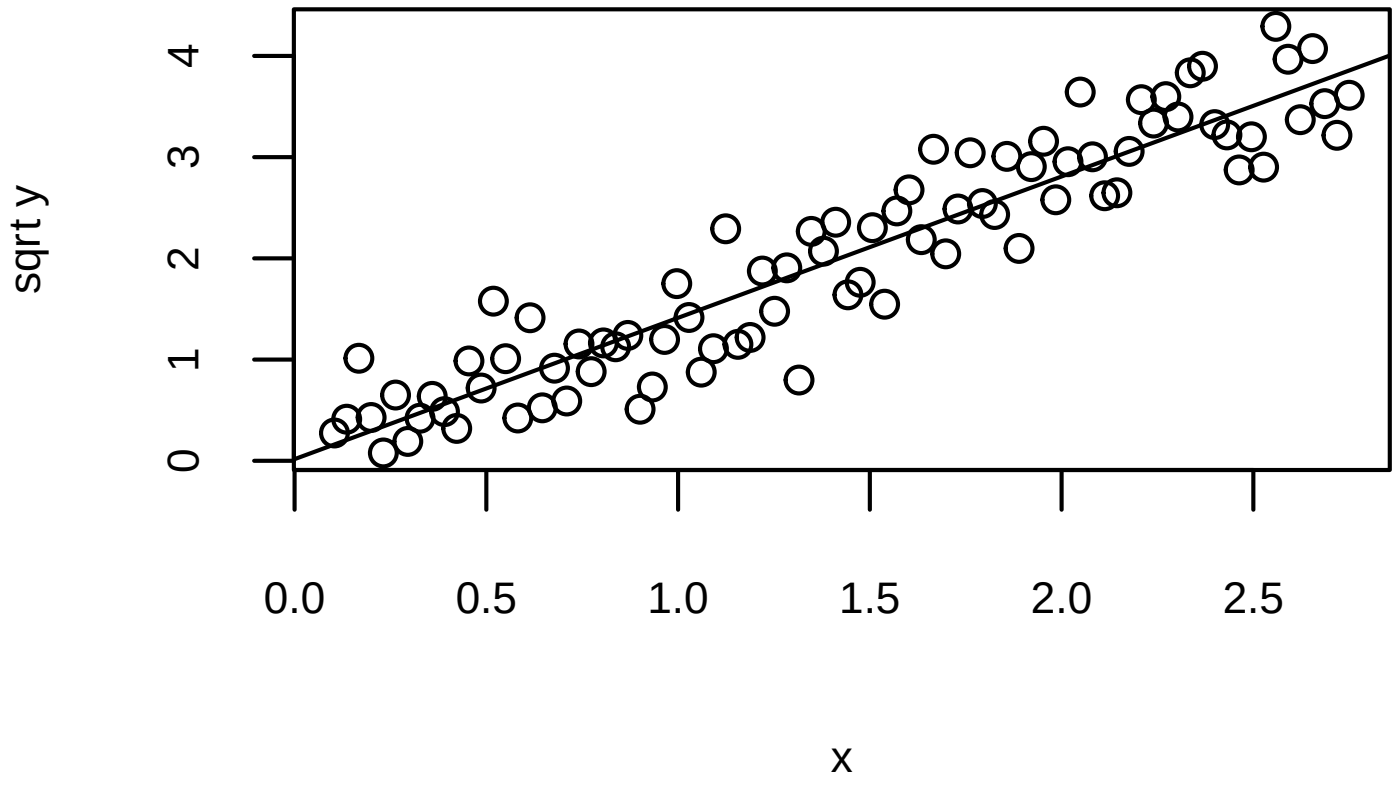
Example 2

What about taking the square root of y?



This actually does look more like a linear relationship. Let's try fitting a model to x and sqrt(y).

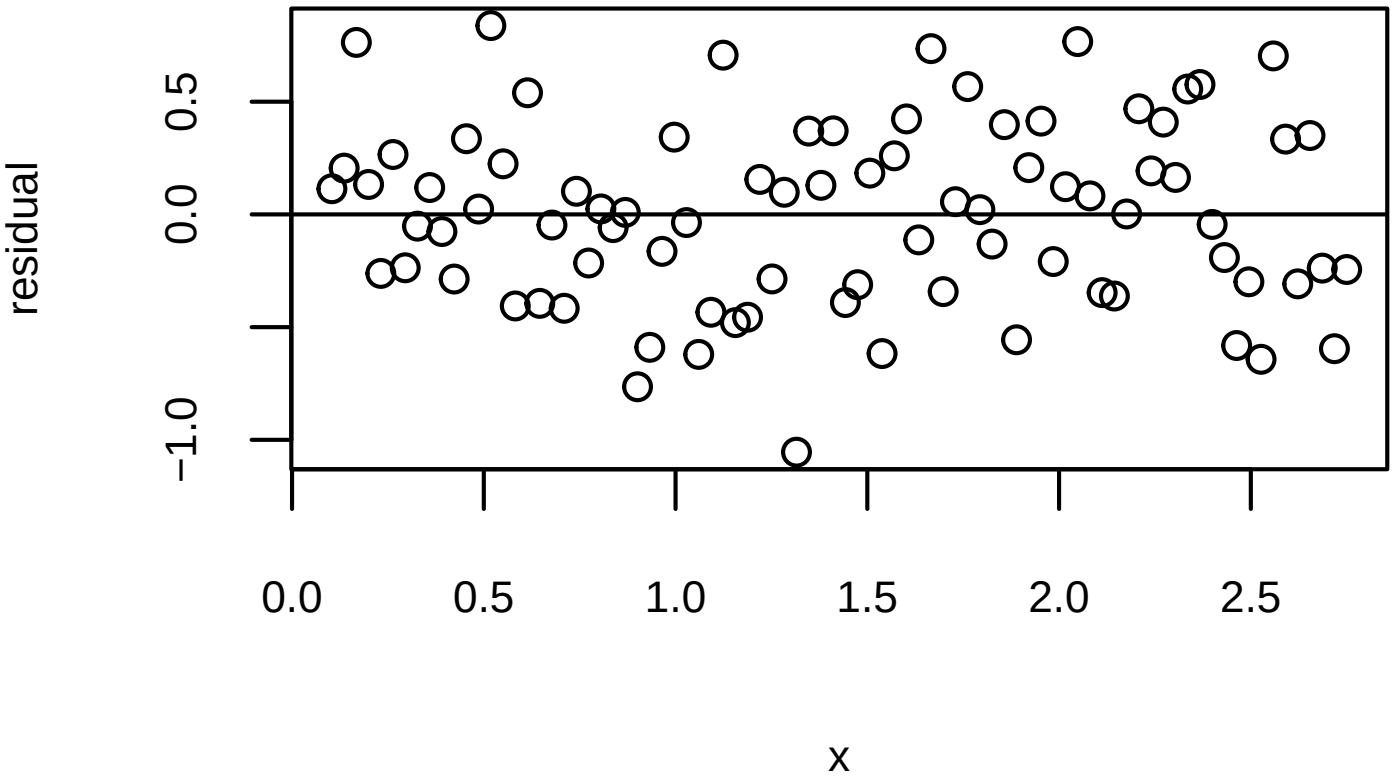
Example 2



(linear regression for x and sqrt(y))

Example 2

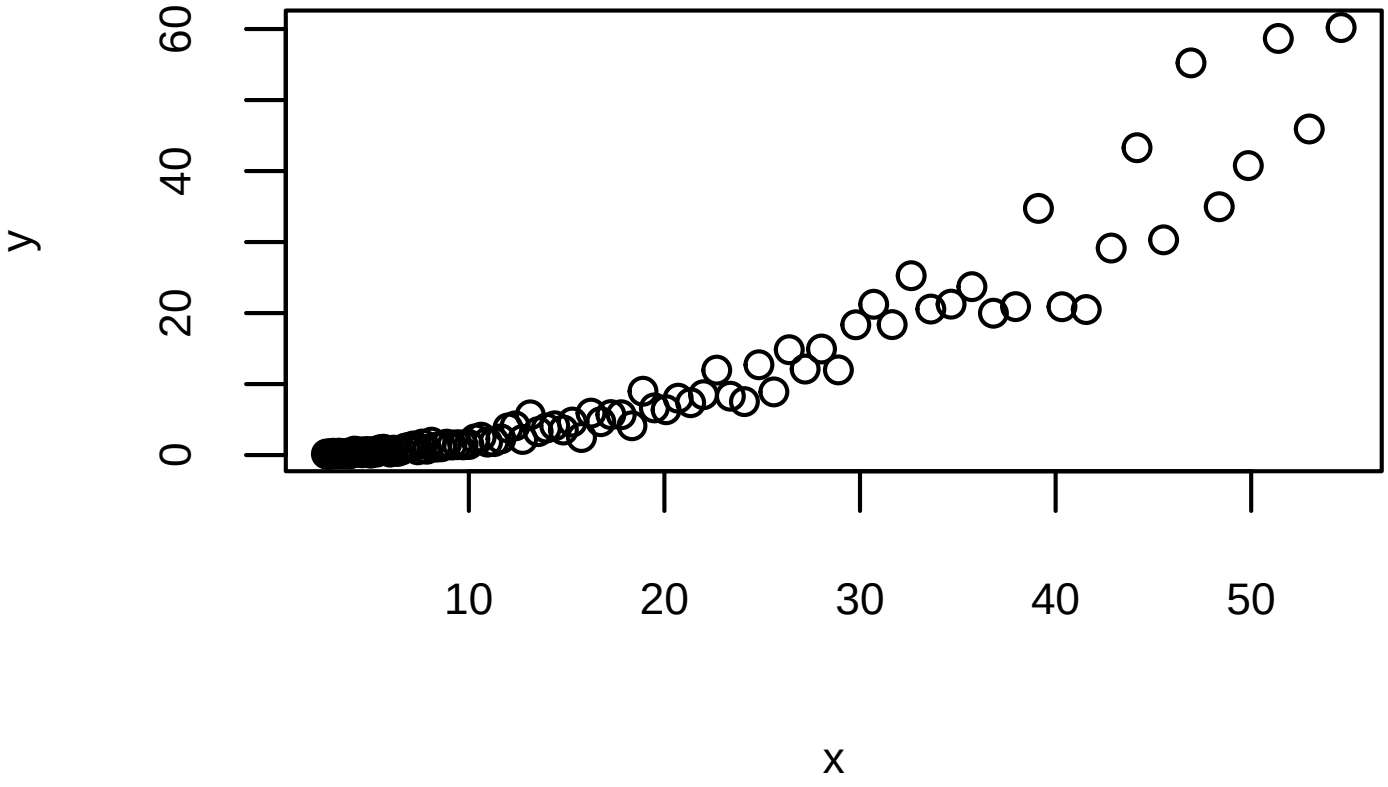
How does the plot of the residuals look now?



The residuals now look more evenly scattered around zero. There is less of an obvious pattern in the residuals based on the value of x. This is satisfactory.

Example 3

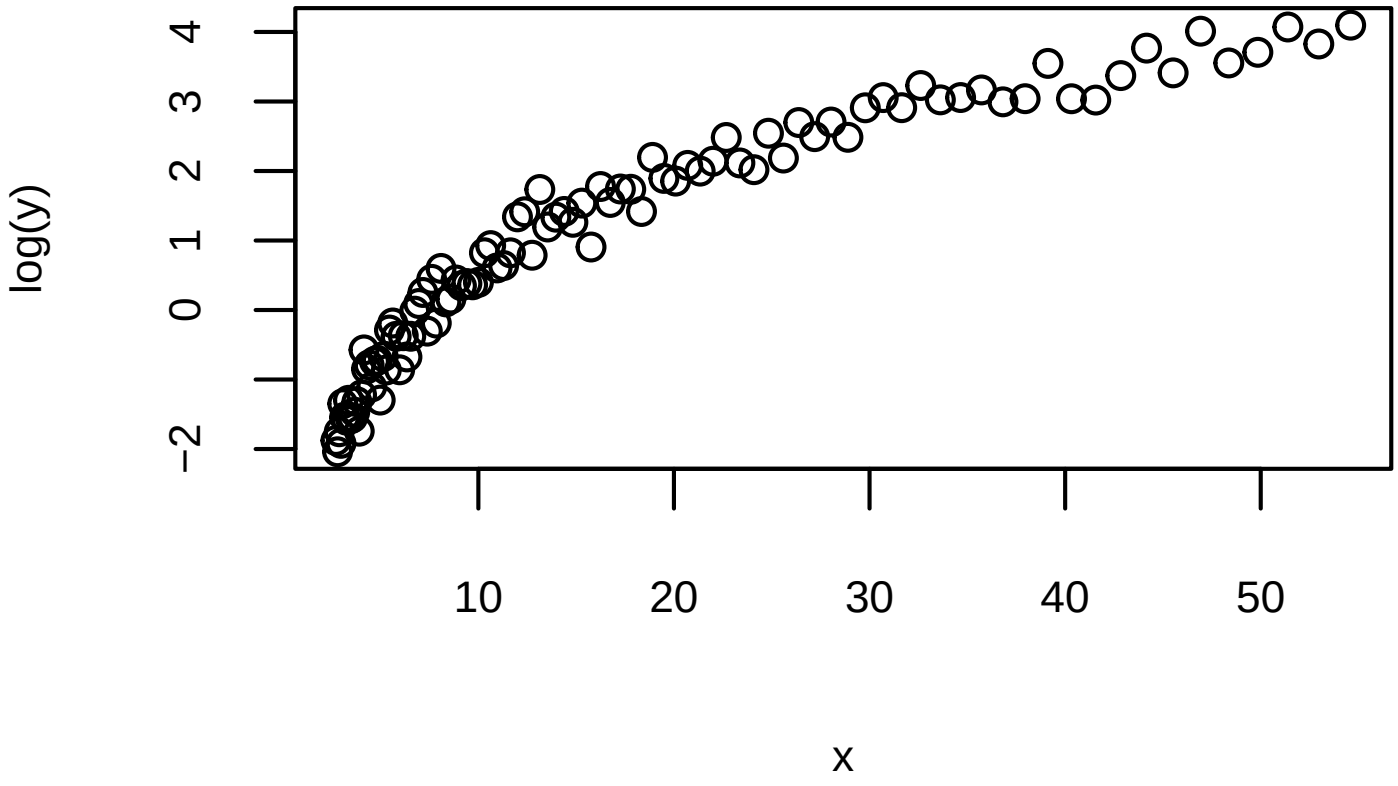
Example 3



Non-linear relationship between x and y.

Example 3

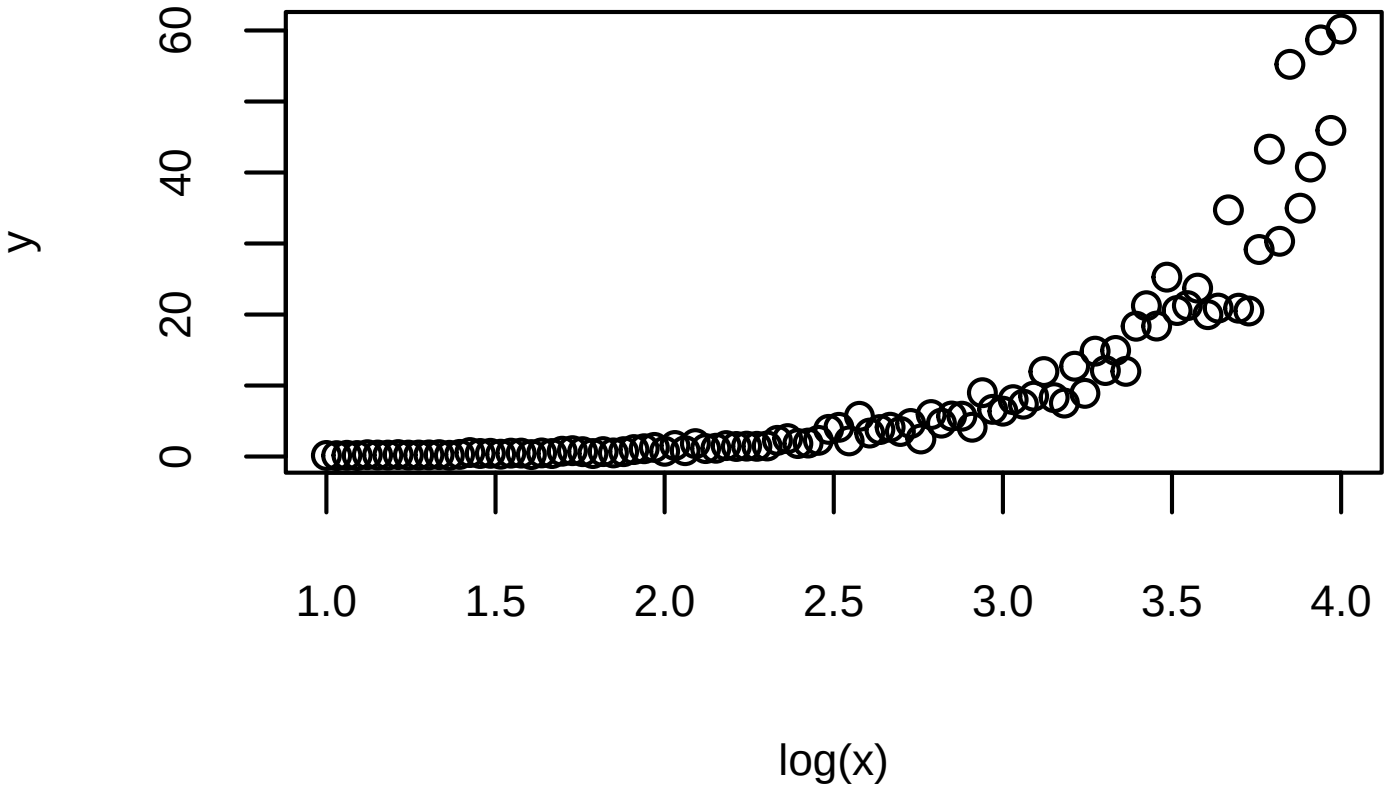
Try replacing y with log(y):



Relationship still non-linear.

Example 3

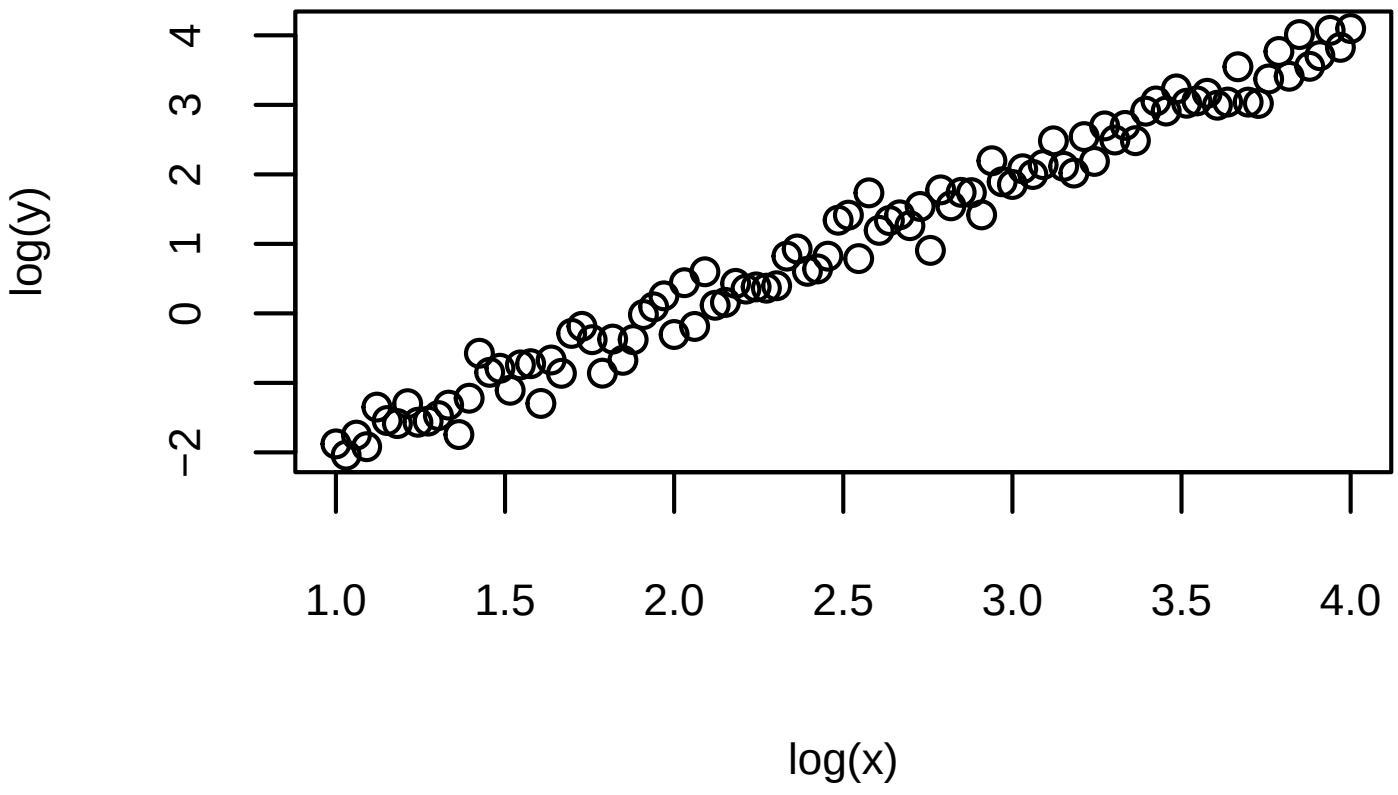
Try replacing x with log(x):



Still non-linear.

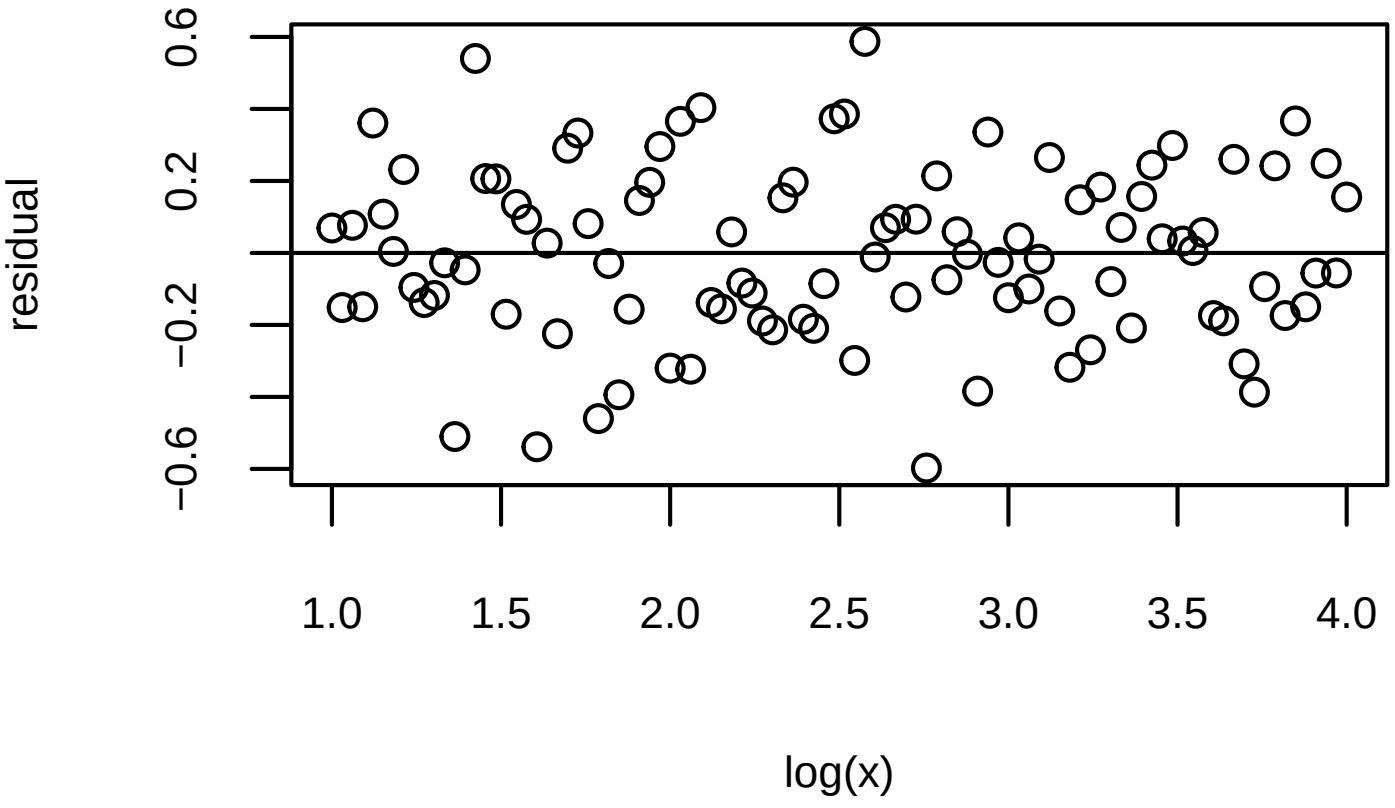
Example 3

Try log(x) and log(y)



This looks better. Try fitting a linear model to log(x) and log(y).

Example 3

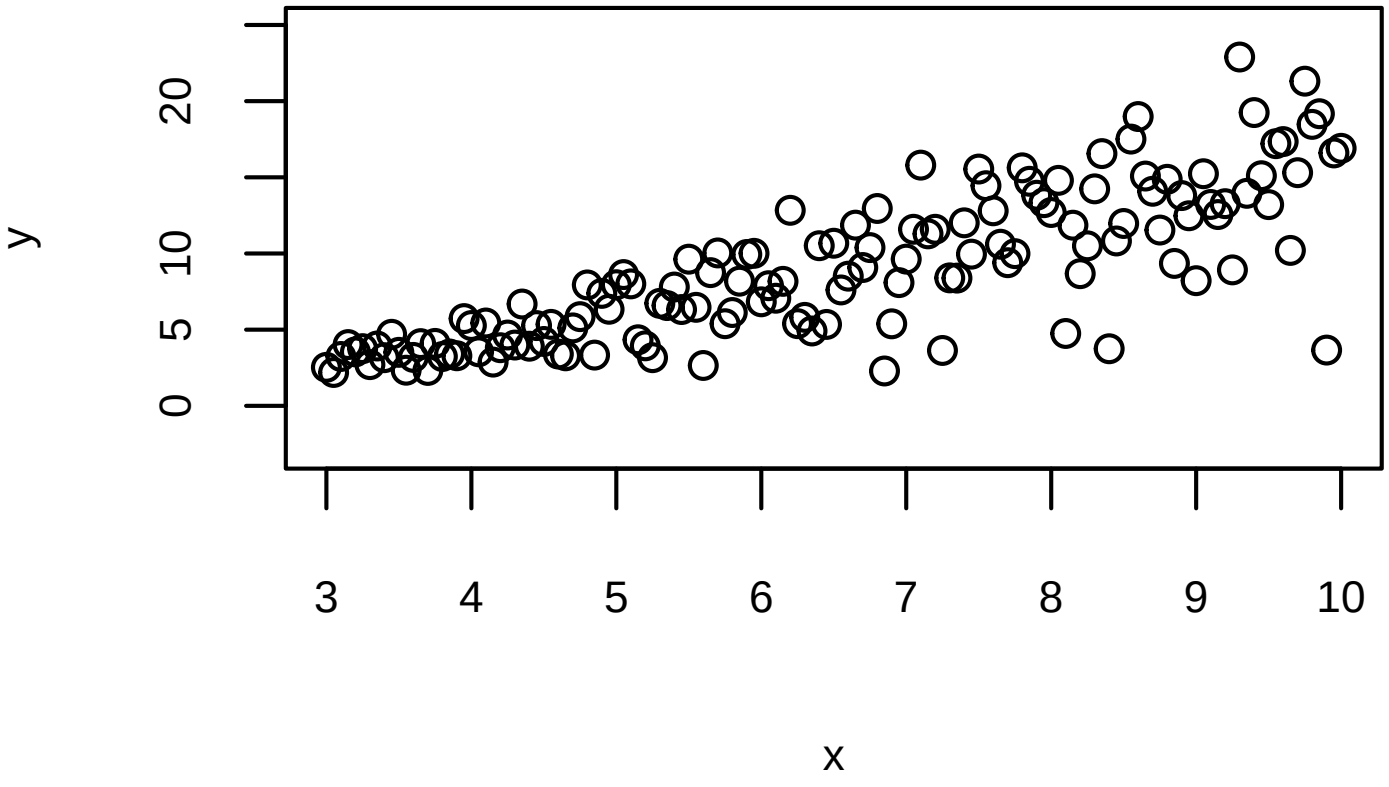


The residuals appear to be evenly scattered around zero, with no clear pattern. Nothing in this plot is concerning.

Example 4

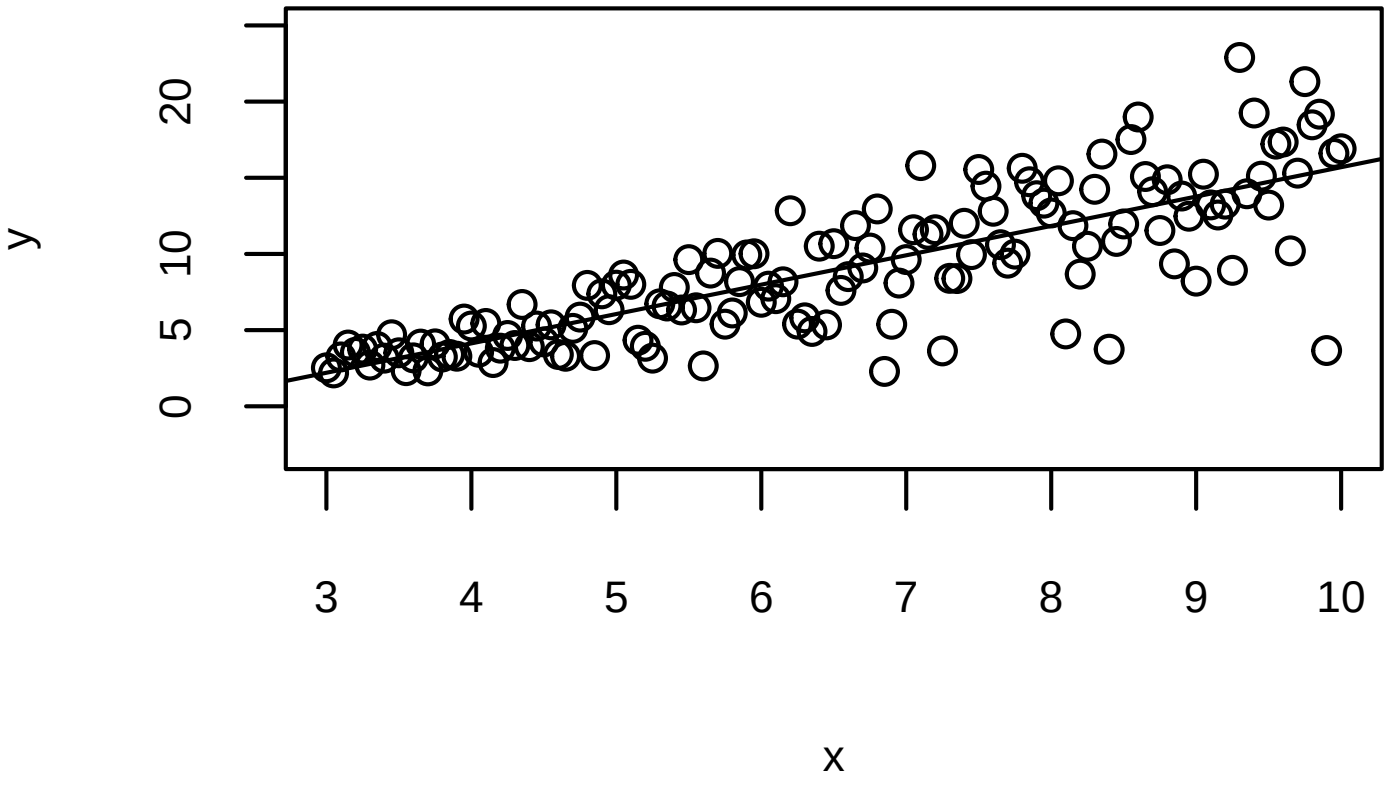
Example 4

Start by looking at a scatter plot of the data, x and y.



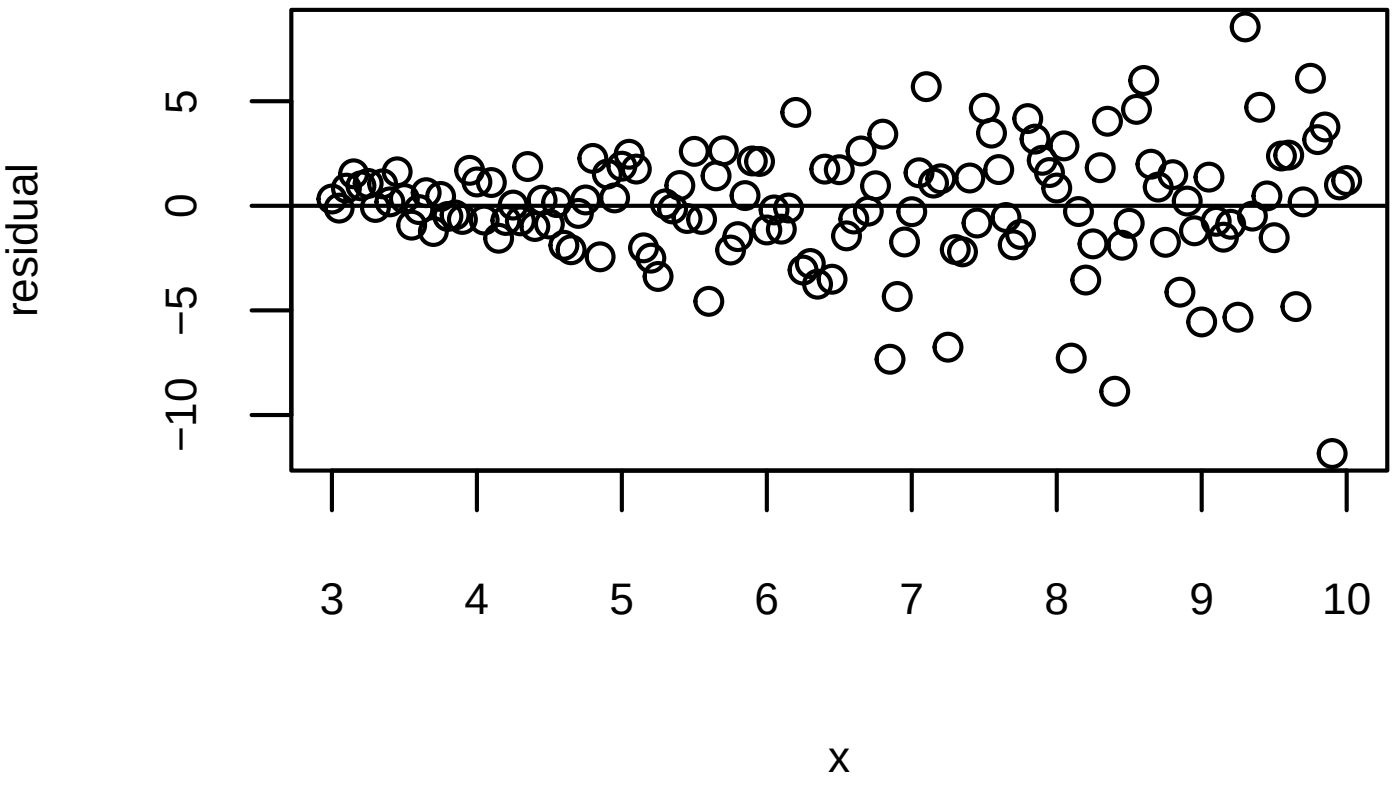
It seems like the relationship between x and y IS linear, but the y values are more spread out at larger values of x.

Example 4



Example 4

What does a plot of the residuals look like for the linear regression model?



There appear to be a roughly equal number of positive and negative residuals at any x value, but the residuals tend to be larger in magnitude for larger values of x.

This suggests the error terms in the true model do NOT all have the same variance.

Example 4

Transforming the x or y variables will typically not be useful in cases like this, because the relationship between x and y already appears to be linear.

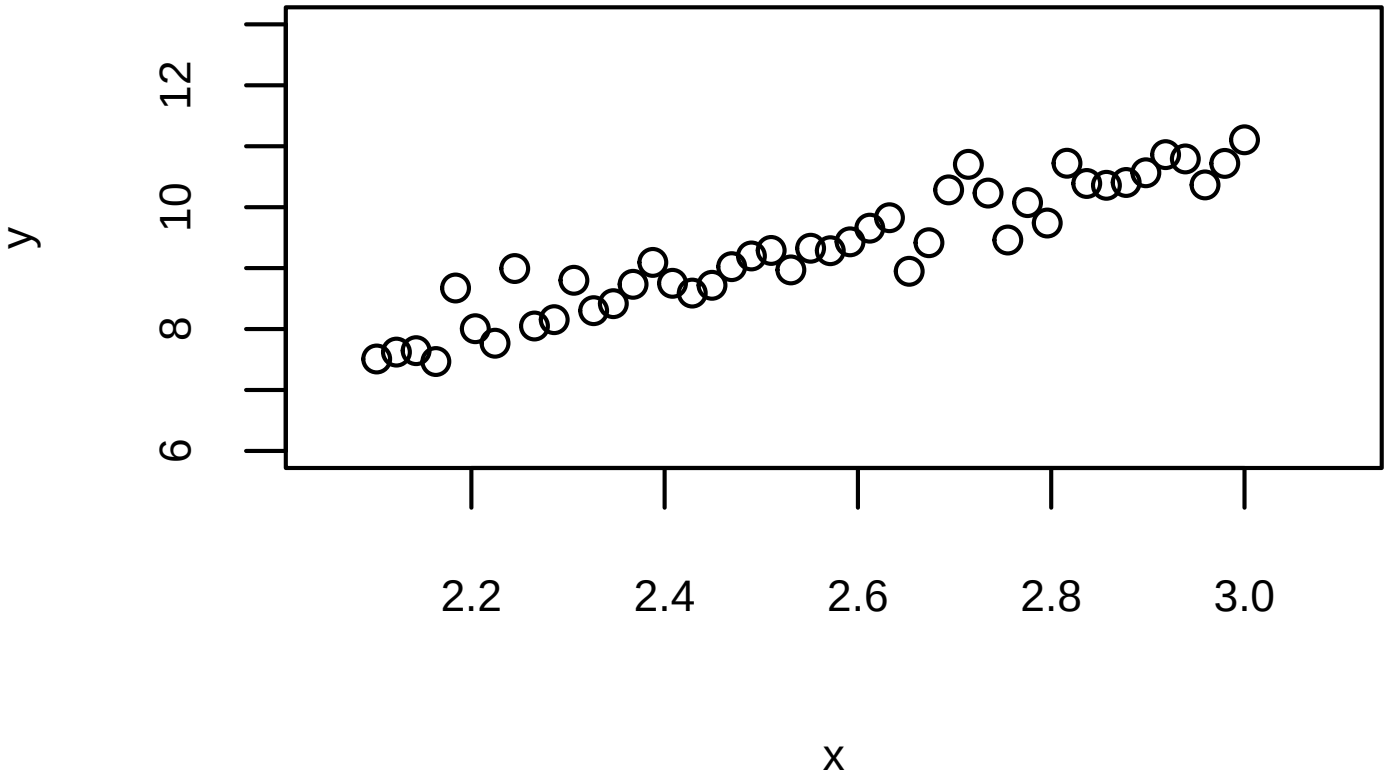
What can we do in this case?

- Still ok to use least squares estimates of slope and intercept, and use those to make predictions.
- But we shouldn't use the usual confidence intervals and hypothesis tests (for any aspect of the model).

Example 5

Example 5

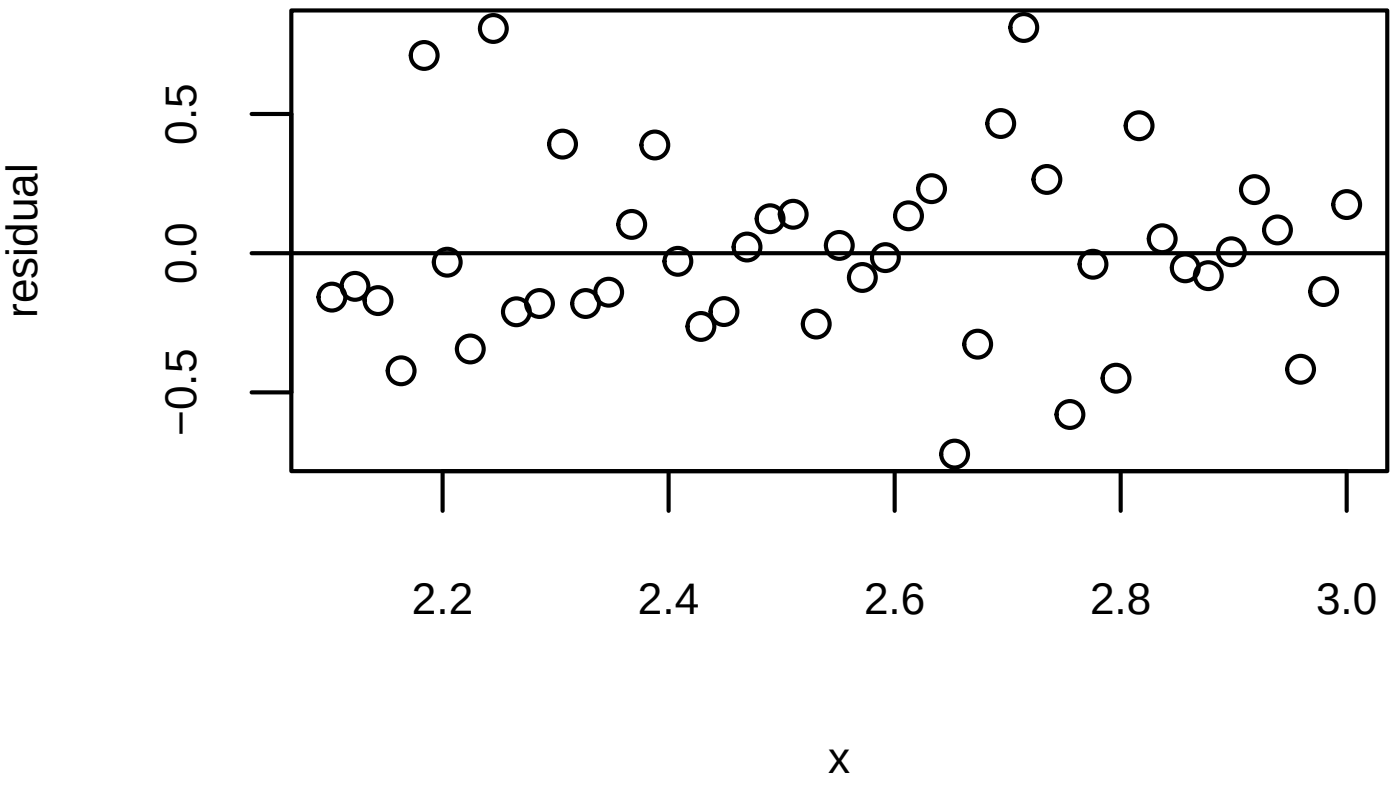
Start by looking at a scatter plot of the data, x and y.



The relationship between x and y looks roughly linear.

Example 5

If we fit a linear regression model to x and y, what do the residuals look like?

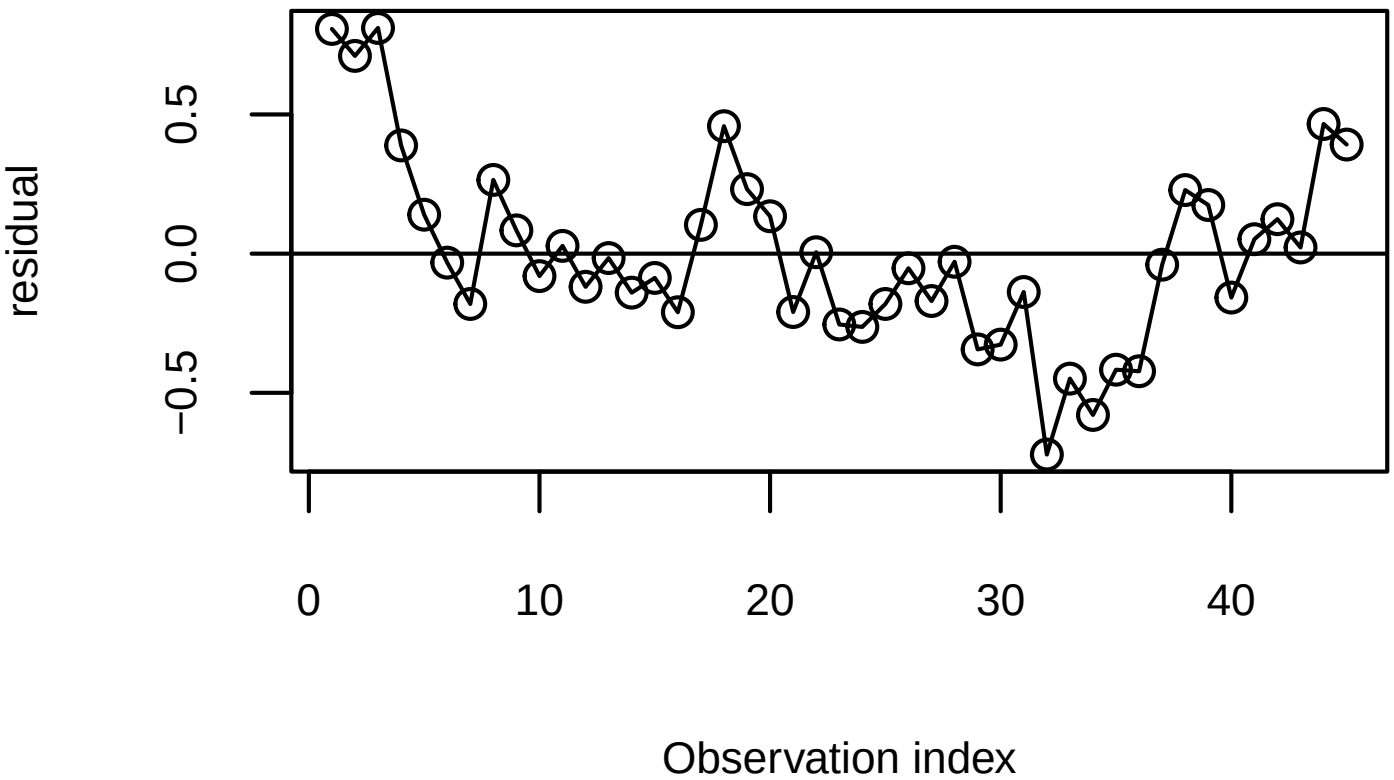


The residuals look randomly scattered above and below zero.

Example 5

Suppose the data consists of sequential observations, e.g. measurements taken in order over time.

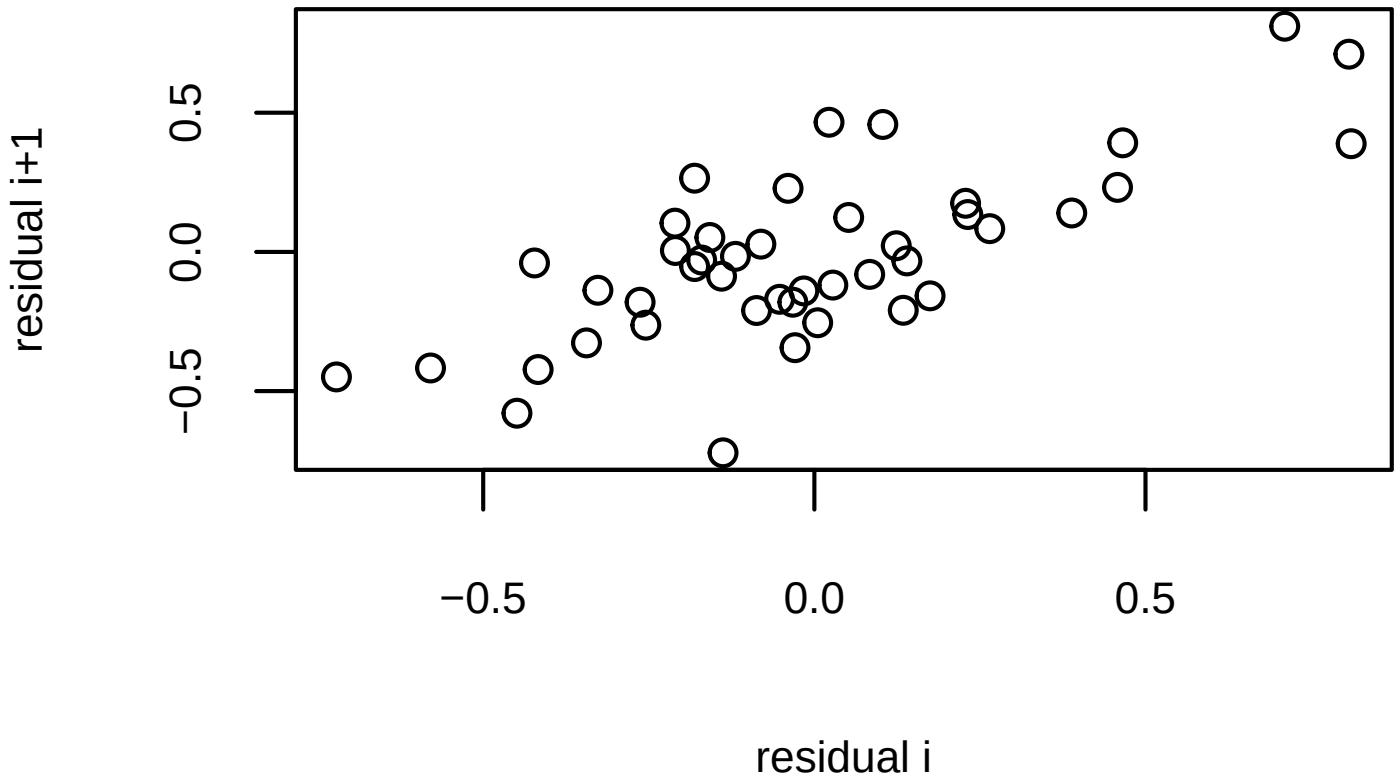
How do the residuals look, plotted against the index? If each data point (x_i, y_i) is a measurement from day i, we may be interested in plotting the residuals with i itself (the index) on the x axis.



We tend to get clusters of several positive residuals in a row, or several negative residuals in a row. This suggests that the errors may be positively correlated.

Example 5

We could also look at a plot of the lagged residuals. On the x axis we put the residual for day i, and on the y axis the residual for the following day. If the values had no association, this plot would look like a symmetric round blob.



This plot supports our suspicion that the error terms may be correlated.

Example 5

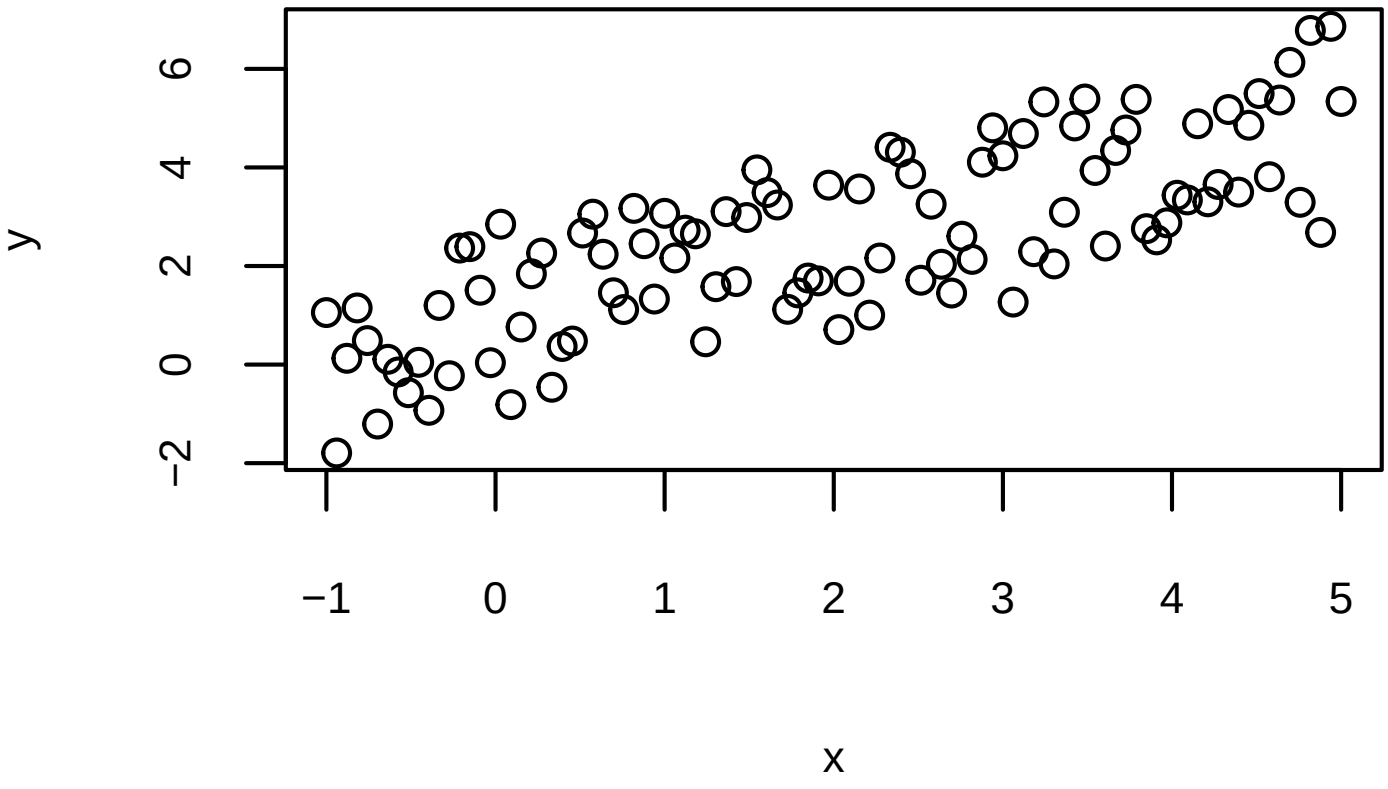
What are the implications?

- The estimated slope and intercept are still valid, and using the fitted regression line to make predictions is also acceptable.
- Confidence intervals and hypothesis tests related to the model (coefficients, predictions) are not correct.
- Alternative types of models exist for this situation, but we have not learned about them yet.

Example 6

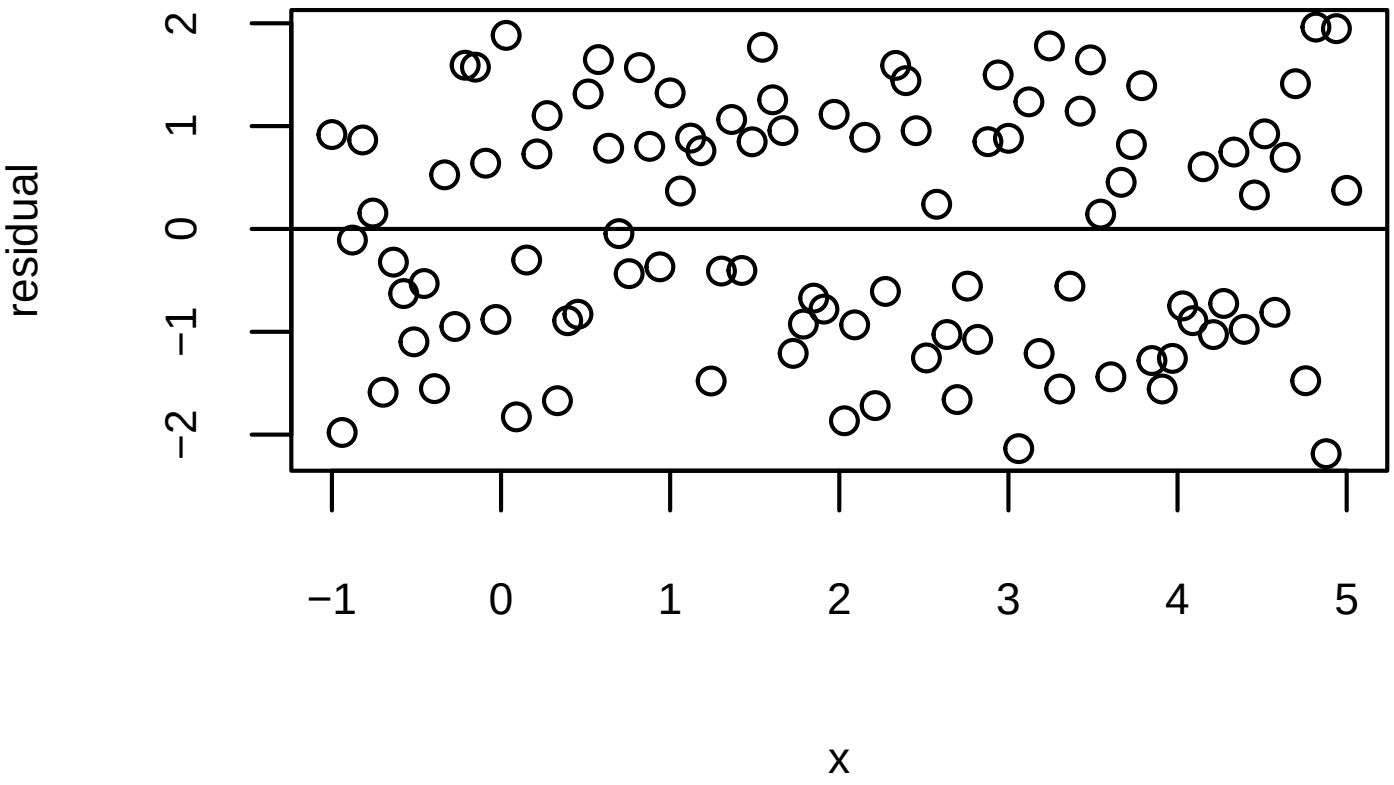
Example 6

Start by looking at a scatter plot of the data, x and y.



The relationship between x and y looks roughly linear.

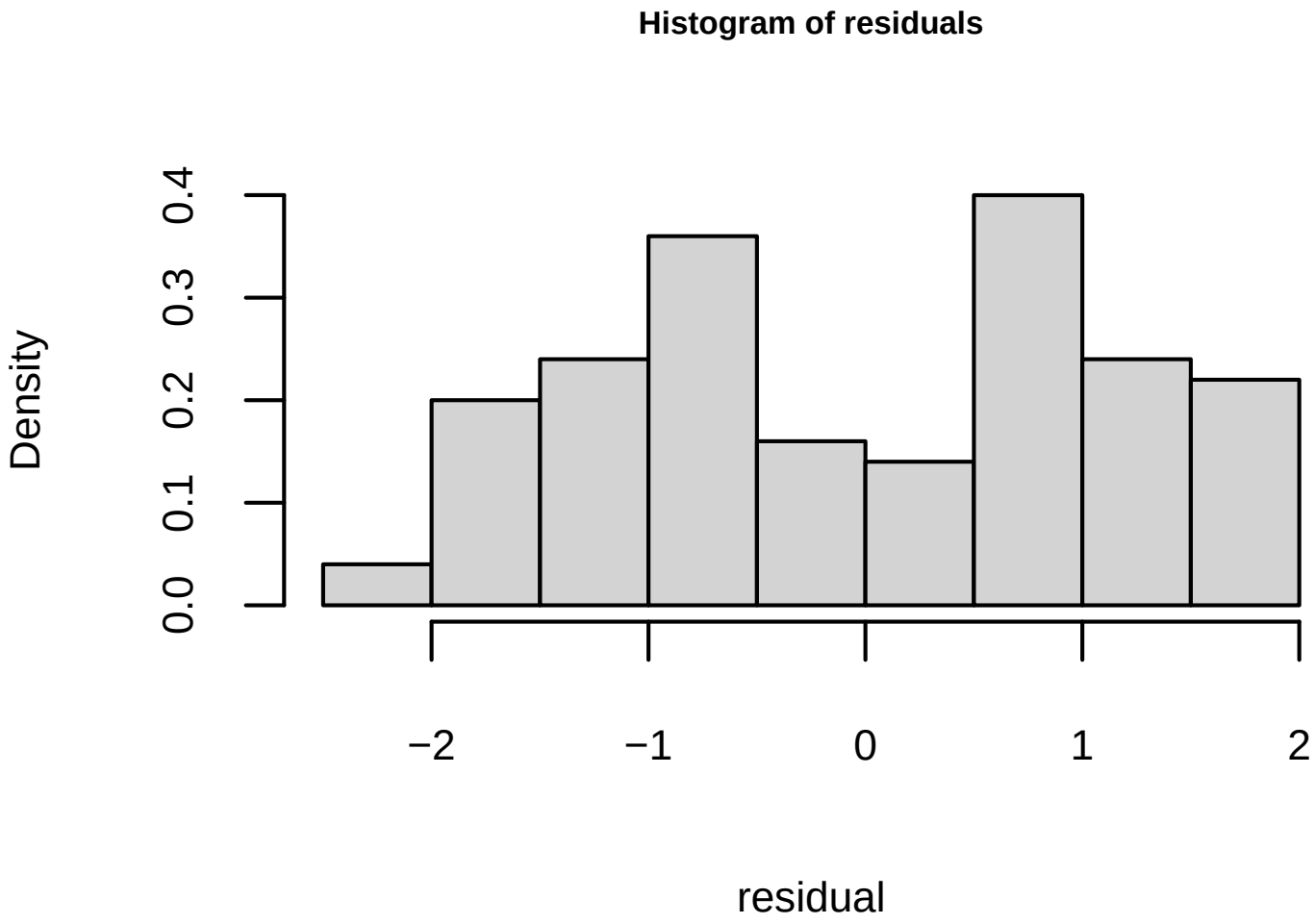
Example 6



The residuals from the fitted model seem to be evenly spread above and below zero, and are not obviously different in magnitude for different values of x.

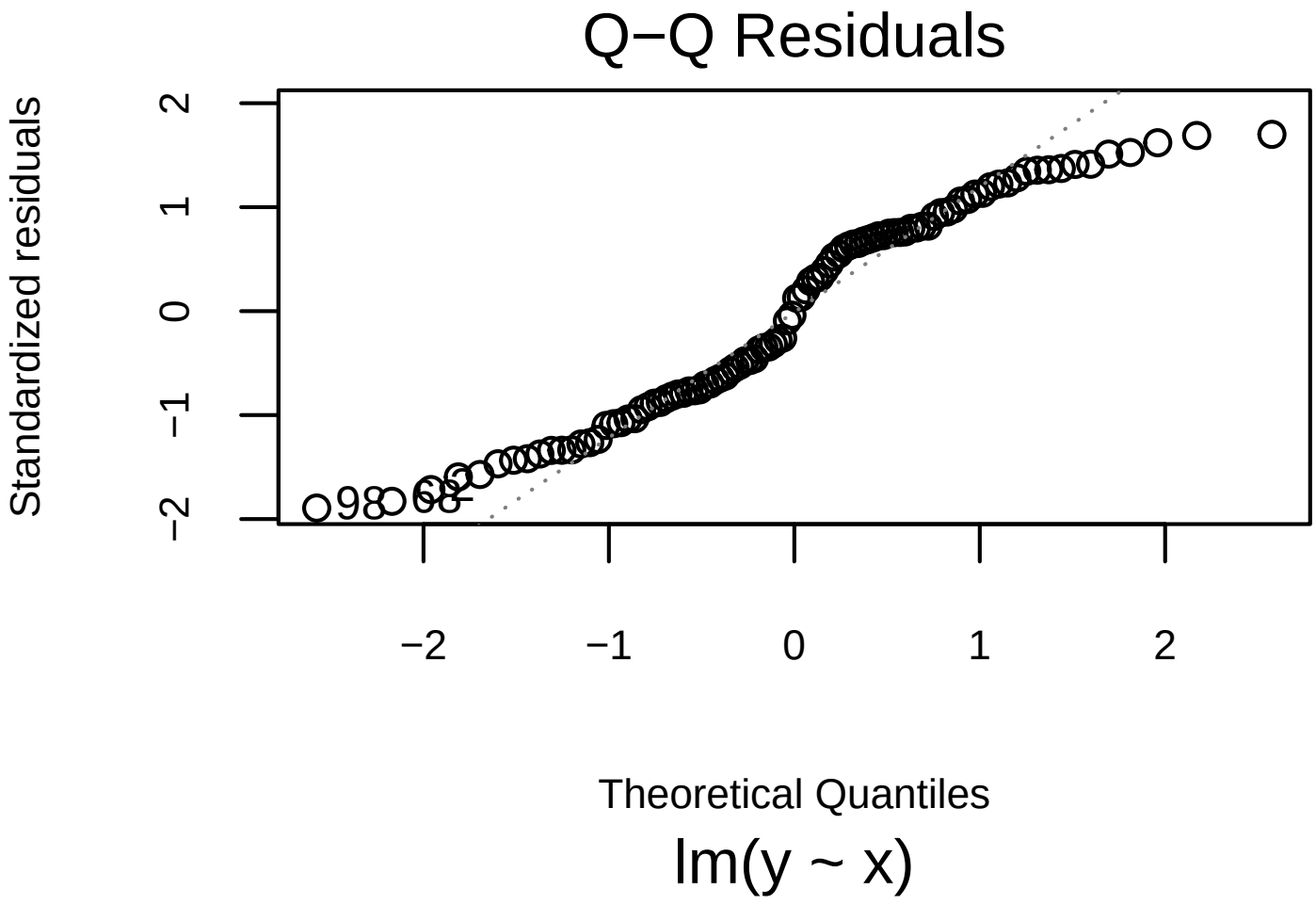
Example 6

However, in this example the residuals do not look like they follow a normal distribution.



Example 6

Q-Q plot can also be used to assess normality



Points on the Q-Q plot deviate from a straight line, which indicates possible violation of normality assumption.

Example 6

What are the implications?

- Estimated slope and intercept are still valid, and using the fitted regression line to make predictions is also acceptable.
- Technically, confidence intervals and hypothesis tests related to the model are not correct...
- ...but in practice, everything **except** prediction intervals for a new y value are probably close to correct.